

# **The role of non-scientific factors vis-à-vis the quality of publications in determining their scholarly impact**

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## **Abstract**

In the evaluation of scientific publications' impact, the interplay between intrinsic quality and non-scientific factors remains a subject of debate. While peer review traditionally assesses quality, bibliometric techniques gauge scholarly impact. This study investigates the role of non-scientific attributes alongside quality scores from peer review in determining scholarly impact. Leveraging data from the first Italian Research Assessment Exercise (VTR 2001-2003) and Web of Science citations, we analyse the relationship between quality scores, non-scientific factors, and publication short- and long-term impact. Our findings shed light on the significance of non-scientific elements overlooked in peer review, offering policymakers and research management insights in choosing evaluation methodologies. Sections delve into the debate, identify non-scientific influences, detail methodologies, present results, and discuss implications.

## **Keywords**

*Research quality; citation-based metrics; non-scientific features; peer review; research evaluation; Italian VTR*

## 1. Introduction

In the realm of equivalent cost, does the superior quality product emerge as the top-selling among those fulfilling a specific need? Probably, but not necessarily. Is it solely the product quality that influences consumer choices? Certainly not. The service integrated into or associated with the product, packaging design, distribution channel, promotion, and general marketing are factors that can contribute to enhancing the product's value and/or the consumer's perception of it. The ultimate goal for a company seeking to maximise profits for its shareholders is not only to produce goods or services better than competitors but to sell more (at an equivalent cost). Therefore, the company also invests in activities complementary to those typically aimed at increasing the intrinsic quality of the product.

With the necessary modifications, the taxpayer, i.e., the shareholder of public research institutions (PROs), and consequently, the policymaker and top management overseeing them, should aim at maximising the socio-economic impact of research expenditures rather than solely focusing on the quality of research output. Notably, many of these institutions have established industrial liaison and technology licensing offices (equivalent to the marketing and sales functions of private companies operating in the market) to promote cross-sector knowledge transfer (social impact). Researchers increasingly turn to social media to expedite the speed, reach, and significance of disseminating research results within the scientific community (intra-sector knowledge transfer, i.e., scholarly impact). Similar to consumer goods, empirical evidence for scientific research "products" indicates that, in addition to intrinsic quality, various non-scientific factors play a role in determining their value/impact (Mammola, Piano, Doretto, Caprio, & Chamberlain, 2022; Xie et al., 2019; Tahamtan, Safipour Afshar, & Ahamdzadeh, 2016).

If the impact of research is what Public Research Organizations (PROs) should maximise rather than quality, then why resort to evaluation methods and incentivising systems based on the assessment of quality through peer review of scientific publications? The latest UK Research Evaluation Framework (REF) 2021, the current descendant of the Research Assessment Exercise (RAE) and precursor to other RAEs<sup>1</sup> adopted by an increasing number of countries under various names (e.g., ERA in Australia, PBRF in New Zealand, VQR in Italy, etc.), "is the system for assessing the quality of research in Higher Education Institutions (HEIs) in the UK," where "the primary outcome of the (evaluation) panels' work will be an overall quality profile awarded to each submission." The quality of submitted research outputs in terms of their originality, significance, and rigour constitutes 60 per cent of the overall performance score. In comparison, the social impact "underpinned by excellent research conducted in the submitted unit" accounts for 25 per cent. Additionally, the "vitality and sustainability" of the research environment contribute 15 per cent.<sup>2</sup> The introduction of social impact evaluation in RAEs is relatively recent and conducted through the analysis of a very limited number of case studies. In any case, the assessment of the quality of research output continues to play a primary role both in the final evaluation and in the performance-based allocation of resources to institutions. These exercises often use a combination of methods, but peer review plays a

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<sup>1</sup> Henceforth, we will use the acronym 'RAEs' to generically refer to national research assessment exercises.

<sup>2</sup> The above phrases in quotes are extracted from REF 2021 – Panel criteria and working methods, <https://www.ref.ac.uk/publications-and-reports/panel-criteria-and-working-methods-201902/>

central role in determining the quality of research outputs and, consequently, the overall research performance of institutions and individuals.

The debate on which of the two approaches is preferable for research evaluation has recently been reignited by the Coalition for Advancing Research Assessment (CoARA) initiative. CoARA advocates that research assessment should be primarily based on qualitative judgment, with peer review playing a central role<sup>3</sup> (Rushworth, 2023; Torres-Salinas, Arroyo-Machado, & Robinson-Garcia, 2023). It is unequivocal that both approaches have pros and cons. Still, it is important to acknowledge that the two methods measure different attributes of research one being the quality of scientific output and the other its scholarly impact. The potential difference between the two should be determined by non-scientific factors associated with the publication, which will be the subject of the present study.

This study aims to ascertain to what extent non-scientific factors contribute to determining, in addition to intrinsic quality, the scholarly impact of a research product. To achieve this, we assume that peer review competently measures quality, and citation-based metrics measure the scholarly impact of research products despite the respective limitations of the two methodologies extensively dissected in the literature (Gingras, 2016; Lee, Sugimoto, Zhang, & Cronin, 2013).

We leverage the knowledge of the quality scores attributed by reviewers to the publications submitted for evaluation in the first Italian RAE, named VTR 2001-2003. For each such publication indexed in the Web of Science (WoS), we measure its citation-based impact. Subsequently, we identify the non-scientific attributes of each publication. Finally, we fit a statistical model to analyse the relationship between the impact of the publication and the quality score assigned by reviewers, controlling for the non-scientific factors. This allows us to gain insight into the weight of non-scientific factors that reviewers may not capture in determining the scholarly impact of a publication.

The results should interest policymakers and research institution management, who must decide whether to measure the quality of research output through peer review or academic impact through bibliometric techniques.

The following section presents the main insights from the debate on peer-review vs bibliometric approaches to assess research. In Section 3, we examine the non-scientific factors that could influence the scholarly impact of publications. In Section 4, we provide details on the data and methods used to measure their relative contribution to impact. In Section 5, we present the results, while in Section 6, we discuss them and draw conclusions.

## **2. Quality or impact, peer review or bibliometrics?**

For a research product to impact the scientific community, it must be utilised (OECD/Eurostat, 2018). To ensure this, it cannot remain tacit but must be encoded in written form beforehand to facilitate its dissemination across all potential users. The most commonly adopted written form by scholars is the article published in scientific journals. The utilisation of the knowledge embedded in the publication is primarily determined by its quality, encompassing originality, significance, and rigour (Jabbour, Jabbour, & de Oliveira, 2013; Patterson & Harris 2009), as well as other non-scientific attributes

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<sup>3</sup> <https://coara.eu/>

(Mammola et al., 2022; Xie et al., 2019; Tahamtan, Safipour Afshar, & Ahamdzadeh, 2016). The contribution of these non-scientific factors may allow a publication of lower quality to have a greater impact than a qualitatively superior one. Therefore, if the intention is to measure impact, evaluating quality alone could lead to distorted results. While bibliometric techniques directly measure a publication's scholarly impact, determined by its quality and non-scientific factors to some extent, the peer-review process is intended to judge its quality.

The pros and cons of each approach have been extensively debated in the literature. The major limitation of peer evaluation is the subjectivity in assessments, as highlighted by the not infrequent disagreements among peers (Schroter, Weber, Loder, Wilkinson, & Kirkham, 2022; Kirman, Simon, & Hays, 2019; Bertocchi, Gambardella, Jappelli, Nappi, & Peracchi, 2015), especially in the case of interdisciplinary works (Thelwall et al., 2023a). Subjectivity occurs not only in the evaluation of the research product but also in the upstream phase of selecting peers (Moxham & Anderson, 1992; Horrobin, 1990) and in the selection of products to be subjected to evaluation (Abramo, D'Angelo, & Di Costa, 2014). Other limitations include potential conflicts of interest, the natural tendency to more favourably evaluate authors with a higher reputation or affiliated with more prestigious institutions, the difficulty of contextualising the judgment on the work to be evaluated in the state of the art at the time of execution, the challenge of identifying quality reviewers as the number of works to be evaluated increases (Abramo, D'Angelo, & Viel, 2013), and last but not least, the costs and time involved in large-scale evaluations, RAEs. Notably, RAEs that adopt peer evaluation base the comparative assessment of institutions on a limited number of total research products to reduce costs and time, albeit at the expense of inevitable distortions in rankings (Abramo, D'Angelo, & Viel, 2010).

Citation-based metrics of the scholarly impact of a publication presuppose that a citation represents a recognition of the influence of the cited publication on the citing one (normative theory of citing) (Mulkay, 1976; Bloor, 1976; Merton, 1973). This assumption is strongly opposed by social constructivists who, on the contrary, believe that persuasion is the primary reason for citing, and therefore, citation-based metrics are not suitable for measuring the impact of scientific work (Brooks, 1985; MacRoberts & MacRoberts, 1984; Knorr-Cetina, 1981; Latour & Woolgar, 1979). The literature has extensively discussed this opposition (Tahamtan & Bornmann, 2018). However, the fact that there is a significant correlation between quality as judged by peers and scholarly impact as measured by citation metrics, both perceived and empirically observed (Jabbour, Jabbour, & de Oliveira, 2013; Patterson & Harris 2009), supports the normative theory of citing.

This does not mean that citations as an impact indicator are free from limitations. In fact, citations do not always reflect quality, as a work can be cited to demonstrate its faults rather than its merits (negative citations). However, this rare event generally occurs soon after publication and should not disrupt the analyses (Pendlebury, 2009). Citations can also be manipulated by authors for their own benefit (excessive recourse to self-references and cross-citations) or by editors who, in some cases, exert pressure on authors to cite works already published in their journals to increase the journals' impact indicators (Pichappan & Sarasvady, 2002). Another limitation is the "delayed recognition" phenomenon, which sometimes affects more mature works (Garfield, 1980; van Raan, 2004; Ke, Ferrara, Radicchi, & Flammini, 2015). Although it has been demonstrated that cases of delayed recognition are quite rare, their effects can still be mitigated by introducing other variables beyond early citations that improve the predictive power of